

Bonus 1 - řešení

$$\textcircled{1} \quad y' = \frac{x^3+2}{y} \quad , \quad y' = xy^3 \quad , \quad y' = \sqrt{y-1}$$

$$\textcircled{2} \quad \frac{dy}{dx} = \frac{x+1}{y^2}$$

$$\int y^2 dy = \int (x+1) dx$$

$$\frac{y^3}{3} = \frac{x^2}{2} + x + C$$

$$y(0)=1: \quad \frac{1}{3} = C \Rightarrow \frac{y^3}{3} = \frac{x^2}{2} + x + \frac{1}{3}$$

$$y = \sqrt[3]{\frac{3x^2}{2} + 3x + 1}$$

$$\textcircled{3} \quad y = C \cdot e^{\frac{x^3}{3} + x}$$